

FCC 3101 CHAPTER 2

HISTORY OF CIVIL ENGINEERING

PART 2

2.7 MODERN AGES

- 2.7.1 The advances in **modern science** during the 17th century, the **industrial revolution** of the 18th century and the proliferation of **new technologies** in the 19th century are the key factors that propelled civil engineering from its humble beginning to its current status.
- 2.7.2 This section outlines some of the key advances and the pioneering scientists and engineers.

2.7.1 Seventeenth Century

- 2.7.1.1 **Modern science** started to have a profound effect on construction. Architect-engineers began to use experimental science to design buildings.
- 2.7.1.2 Construction tools, construction materials and construction techniques improved considerably.
- 2.7.1.3 **Galileo Galilei** (1564 – 1642):
An Italian philosopher, who is considered to be the father of modern science, made outstanding contributions to **science, astronomy and engineering.**

2.7.1.4 In 1638, **while under house arrest**, **Galileo** published his “***Two New Sciences***” where he laid down the foundations for **Kinematics of Objects** and **Strength of Materials**. That was the **true beginning of engineering**.

2.7.1.5 **Robert Hooke** (1635 – 1703):
An English polymath is credited with formulating **Hooke’s law of linear elasticity** in 1676.

2.7.2 Eighteenth Century

- 2.7.2.1 The onset of the **industrial revolution** in **1760** saw a significant increase in the size of cities. Industries were built to manufacture virtually everything.
- 2.7.2.2 Construction started to be **mechanized**. There were many technological advancements.
- 2.7.2.3 **Architecture** and **engineering** became two distinct professions in **1750**.
- 2.7.2.4 Experimental science and mathematical analysis became the basis of civil engineering design.

2.7.2.5 Many people made significant contributions in science, astronomy, mathematics and engineering. Some of the **pioneer contributors** in civil engineering are acknowledged below: -

- 1. Isaac Newton (1642 – 1727):** An English polymath who is credited with formulating **Newton's laws of motion**.
- 2. Johann Bernoulli (1667 – 1748):** A Swiss mathematician who is credited with systematizing the **principle of virtual work**.

3. **Leonhard Euler** (1707 – 1783): A Swiss mathematician who is credited (together with **Daniel Bernoulli**) with deriving the classical **beam deflection equation** and **Euler's critical load for strut buckling**.
4. **Daniel Bernoulli** (1700 – 1782): A Swiss mathematician (who in addition to working with **Euler**), is credited with formulating the **Bernoulli principle in fluid mechanics**.

2.7.3 Nineteenth Century

- 2.7.3.1 By the beginning of the nineteenth century, industrial revolution had spread to many countries in western Europe.
- 2.7.3.2 New construction works included **railway lines** and **macadam roads**.
- 2.7.3.3 New technologies included **steam engines**, **machine tools** and **explosives** to aid in mining.
- 2.7.3.4 The **circular saw**, **steel nails**, **rivets** and **screws** were invented. **Water supply** systems and **waste water collection** and **treatment** came into use.

- 2.7.3.5 In 1824, **Joseph Aspdin** developed a method for producing **Portland cement**.
- 2.7.3.6 In 1849, **Joseph Monier** invented **reinforced concrete**.
- 2.7.3.7 In 1850, mass production and processing of **steel beams** and **columns** started.
- 2.7.3.8 Concrete **reinforcing bars** and **glass panes** also went into mass production.
- 2.7.3.9 **Design** and construction of buildings and other structures became **fully** anchored on **scientific** knowledge and **mathematical** tools for analysis.

2.7.3.10

Many people made significant **contributions in civil engineering.**

2.7.3.11

Below is a list of some of the contributors whose work has remained **classical**: -

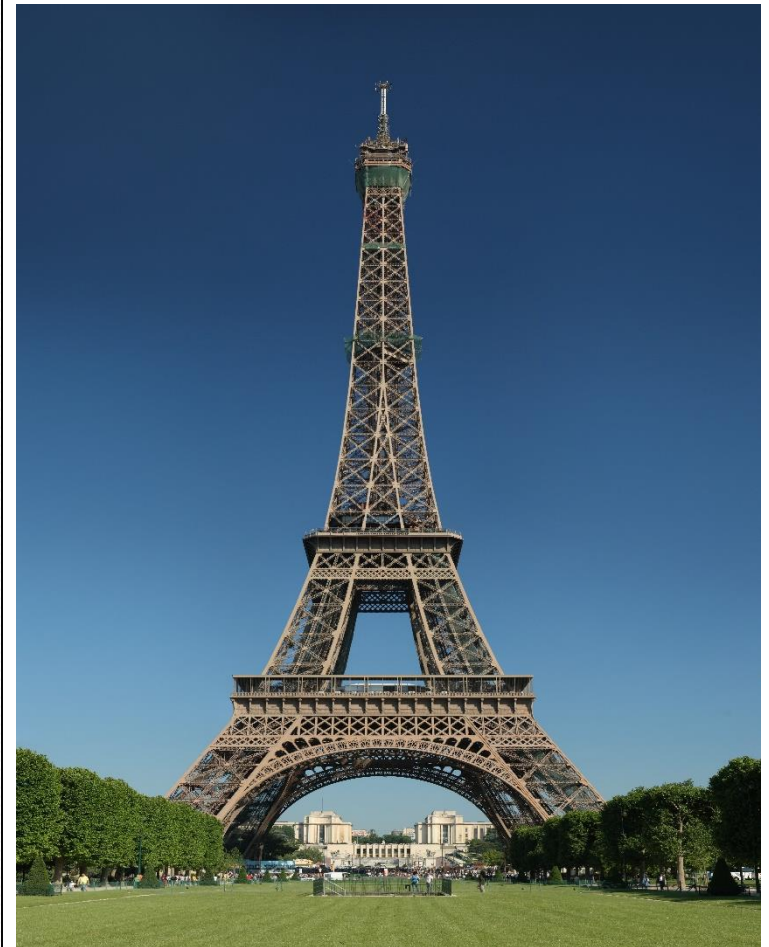


Figure 2.7 Eiffel Tower, Paris 1889

1. Claude-Louis **Navier** (France) published **Elasticity of Structures (1821)**.
2. Benoit Paul Emile **Clapeyron** (France) formulated **Three Moment Equation (1857)**.
3. James Clerk **Maxwell** (Scotland) developed the **Reciprocal Theorem (1861)**.
4. Christian Otto **Mohr** (Germany) developed **Mohr's Analogy (1870)**, **Mohr's Moment Area Theorems (1870)** and **Mohr's Circle of Stresses (1882)**.
5. Carlo Alberto **Castigliano** (Italy) formulated **Castigliano's Two Theorems (1873)**.

6. **Muller-Breslau** (Germany) developed **Influence Lines (1890)** and the **Conjugate Beam method (1895)**.
7. Charles Ezra **Greene** (USA) developed the **Moment Area Method (1897)**.
8. George A **Maney** (USA) developed the **Slope Deflection Method (1914)**.
9. William Herrick **Macaulay** (England) developed **Macaulay's Method (1919)**.
10. Hardy **Cross** (USA) developed the **Moment Distribution Method (1930)**.

2.7.4 Twentieth Century

- 2.7.4.1 **Elevators and cranes** made the construction of **high rise buildings** and **skyscrapers** possible.
- 2.7.4.2 Heavy construction equipment and power tools changed the construction process from **labor intensive** to **capital intensive**.
- 2.7.4.3 **Computer aided design** was introduced, replacing the traditional drawing boards.

- 2.7.4.4 Steel and timber **pre-fabrication technologies** evolved making construction easier.
- 2.7.4.5 **Precast** and **pre-stressed** concrete technologies also evolved making reinforced concrete the most promising structural material for the future.
- 2.7.4.6 **Occupational health and safety, energy conservation and environmental protection** were incorporated into design and construction.

2.8 CURRENT STATUS

- 2.8.1 Currently, civil engineering is **one of the two core disciplines** involved in **infrastructural development**.
- 2.8.2 **Infrastructure** is defined as the physical systems which provide commodities and services which are essential to **sustain** and **enhance** human living conditions, while **preserving** the environment.
- 2.8.3 **Development** is defined as the creation of the infrastructure necessary for **sustainable** and **progressive** socio-economic conditions.

2.8.4 Currently, civil engineering enjoys many **strengths** which include: -

1. Adequate knowledge of **structural behavior**.
2. Adequate knowledge of **structural materials**.
3. Adequate **computational power**.
4. Advanced **construction technologies**.
5. Heavy **construction machinery** and power tools.

2.8.5 However, there are **challenges** facing civil engineering which include: -

1. Imbalances in **levels of economy**.
2. Imbalances in **levels of technology**.
3. Poor **technology transfer protocols**.
4. **Environmental** protection **issues**.
5. **Geopolitical issues**.

That is the History of Civil Engineering